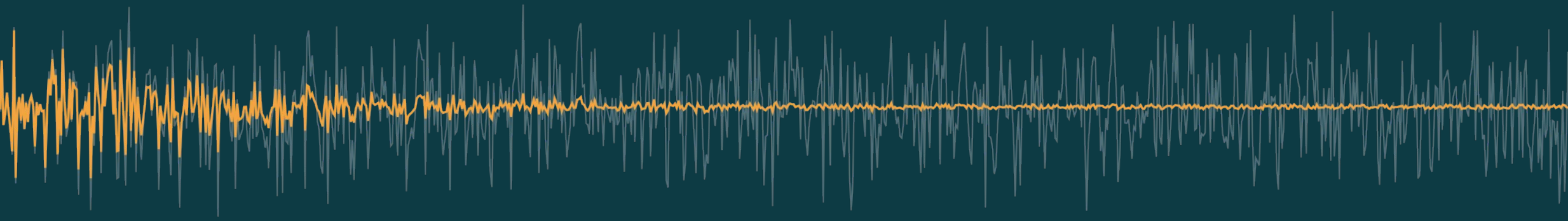
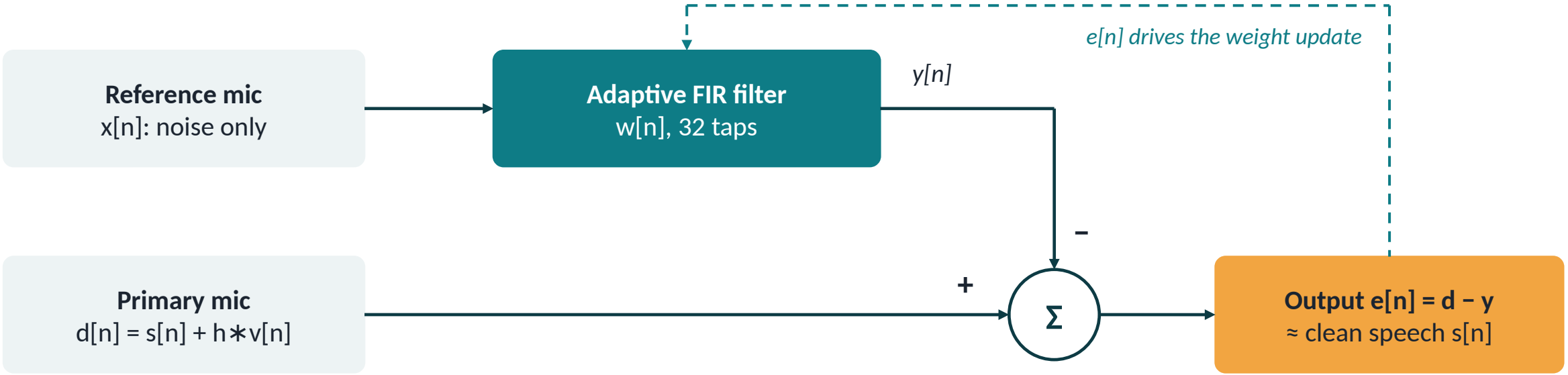
ULLAP

A bit-exact, processor-free FPGA adaptive noise canceller: LMS, NLMS and divider-free NLMS-Shift on the Terasic DE2-115



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Guide: Prof. Jaffino G · SENSE, VIT Vellore · October 2026

Cancelling noise needs a reference and an adaptive filter



Sample by sample, fixed latency

A dedicated FPGA datapath answers 37 clock cycles (0.74 µs) after each input, with no processor or OS.

LMS is cheap but level-sensitive

One fixed step size is too slow when the noise drops or too noisy when it rises.

NLMS needs a division per sample

Normalising by input power fixes the level problem, but a hardware divider costs logic and cycles.

Three algorithms share one update equation

$$y[n] = w^T[n]\,x[n] \qquad e[n] = d[n] - y[n] \qquad w[n+1] = w[n] + \mu[n] \cdot e[n] \cdot x[n]$$

LMS

$$\mu[n] = \mu$$

Fixed step, μ = 0.25

- Smallest: 1,792 logic elements
- 73 busy cycles per sample
- Step is right for only one input level

NLMS

$$\mu[n] = \mu \,/\, (P[n] + \varepsilon)$$

$P[n] = \|x[n]\|^2$, restoring divider

- Level-independent adaptation
- 130 busy cycles per sample
- Divider runs one iteration per bit

NLMS-Shift

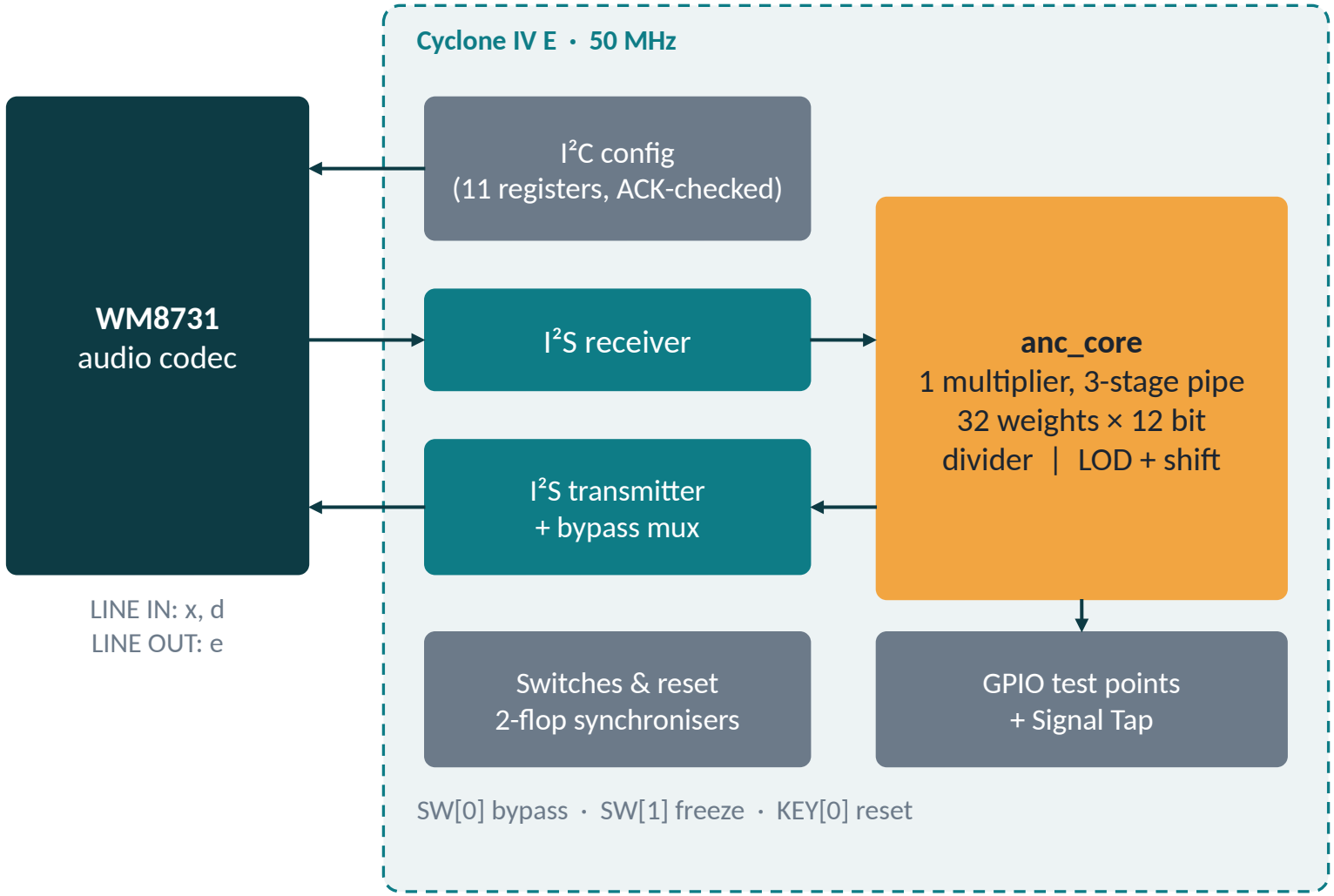
$$\mu[n] = 2^{-(\lfloor \log_2(P+\varepsilon) \rfloor + 1 + K)}$$

Leading-one detector + barrel shift

- Keeps NLMS's level-independence
- 72 busy cycles per sample
- Step within ×0.5 to ×1 of NLMS

Not claimed as new: power-of-two normalisation of NLMS was used by Garrett et al. (ISLPED 2002). Our contribution is the fair comparison, the bit-exact chain to hardware and the analysis.

A complete audio system on the DE2-115, no processor



8.138 kHz

codec master, 12.5 MHz clock, 1536 fs

37 cycles

input-to-output latency (0.74 µs); 35 for LMS

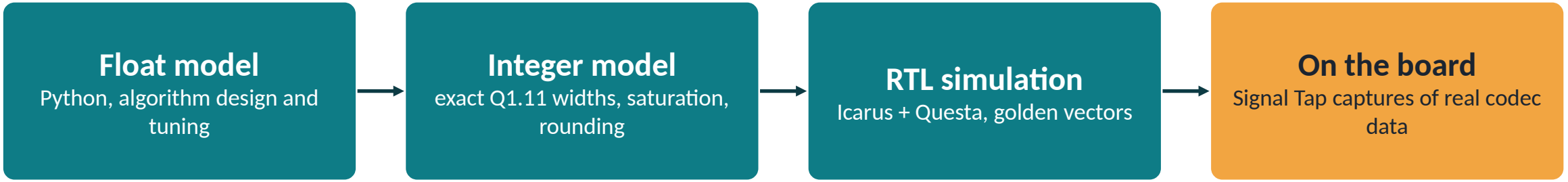
72 / 6,144

busy cycles per sample period for NLMS-Shift: about 1% of the time

0

processors or vendor IP blocks: plain Verilog

One chain of evidence: model, RTL, silicon



Suite	Tests	Result
Core golden vectors (W 8–31, L 2–32, saturation, freeze)	68	all bit-exact
Protocol corner cases	4	pass
Freeze hold / toggle variants	3	pass
Top level with codec model (T1–T10)	11	pass
Python unit tests	53	pass
Verilator lint (-Wall)	–	0 unwaived warnings
On-board Signal Tap captures	**25**	**all bit-exact**
On-board freeze, DAC word, 10-min soak	**4**	**pass**

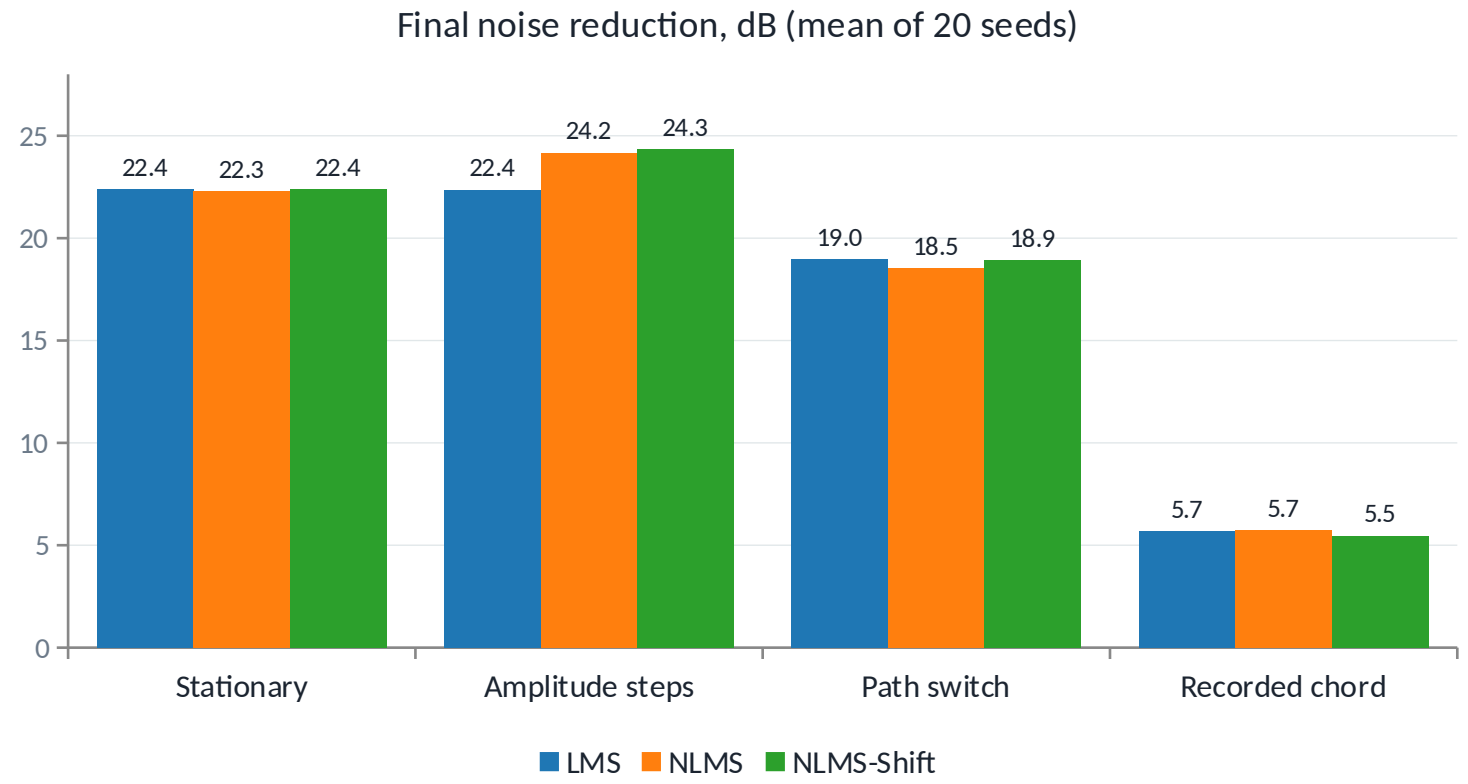
Tests that can fail

Mutation tests inject faults into the RTL and confirm the testbenches catch them, so a pass means something.

Tuned fairly, NLMS-Shift ties NLMS and beats LMS on steps

Set-up

- 4 scenarios, 16,000 samples at 8 kHz: stationary, amplitude steps (×3 then ×0.25), path switch, recorded chord
- Step sizes tuned on seeds 101–103 with one criterion for all
- Reported on seeds 1–20, 95% CI, paired TOST (±0.5 dB)
- LMS μ = 0.25 · NLMS μ = 0.25, ε = 1 · Shift K = 3, ε = 0.25



+1.8 to +2.0 dB for both normalised variants over LMS on amplitude steps ($p < 10^{-7}$). Every NLMS-Shift vs NLMS difference is inside ±0.5 dB.

On real noise and speech, the speech is the limit

MS-SNSD noise, 0 dB SNR	NR (dB)	STOI	PESQ (NB)
Air conditioner	11.7–12.0	0.78 → 0.95	1.54 → 2.30–2.34
Vacuum cleaner	14.3–14.9	0.73 → 0.98	1.39 → 2.81
Copy machine	14.0–14.6	0.69 → 0.98	1.36 → 2.73–2.75
Babble	12.0–12.4	0.72 → 0.96	1.48 → 2.47–2.48

Range over LMS, NLMS and NLMS-Shift; 20 seeds each. NLMS leads LMS by 0.4–0.6 dB, NLMS-Shift sits between them.



Mic placement matters more than the algorithm. With −20 dB of speech leaking into the reference, NR drops 2.3–4.6 dB and PESQ by 0.25–0.6.

Same noise, speech removed

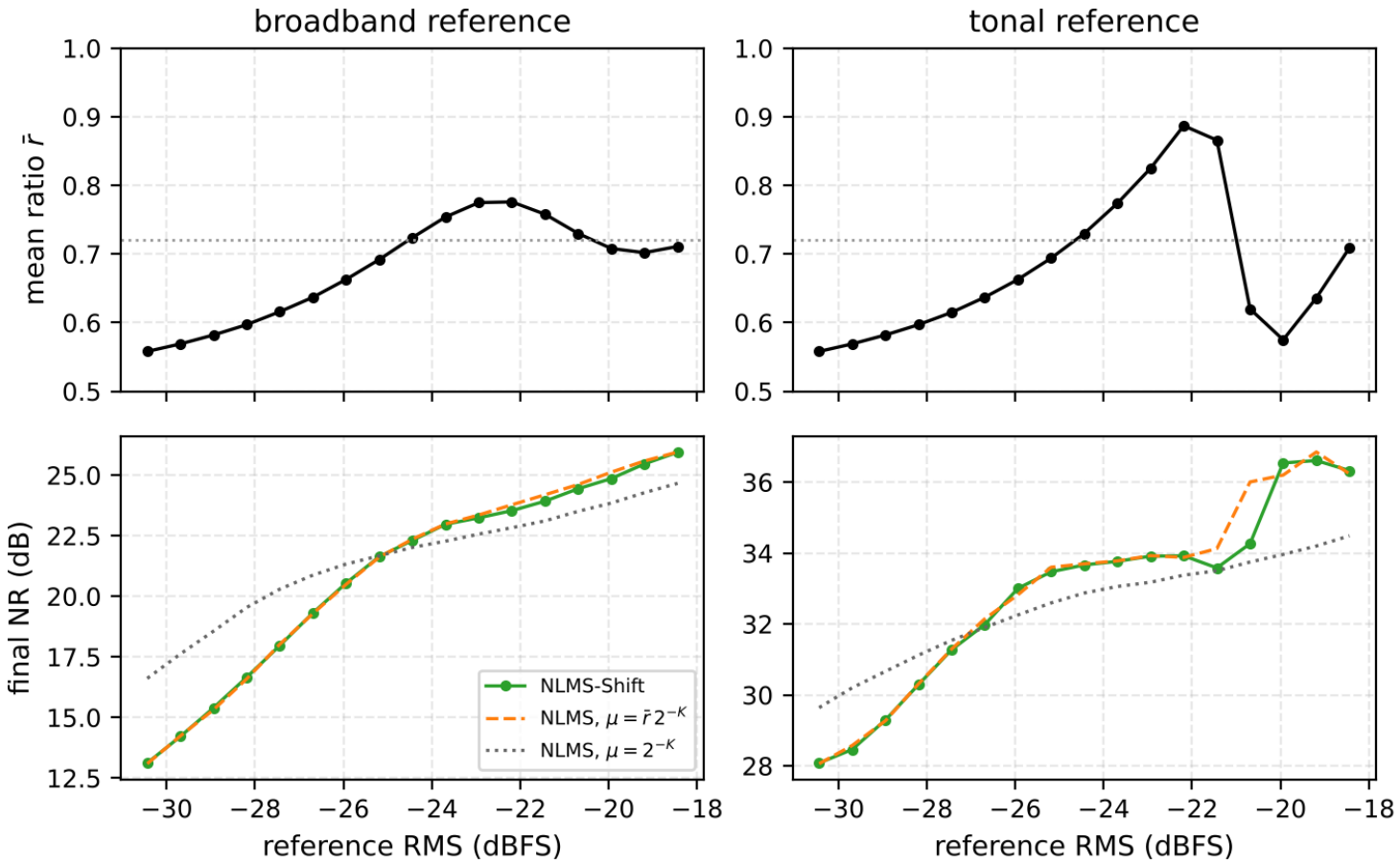
18.5–31.9 dB

With speech

11.7–14.9 dB

Speech enters e[n], which drives the update, and costs 6.4–17.9 dB. Freezing adaptation during speech (a voice-activity detector on the freeze input) is the known remedy; not evaluated here.

What the shift really changes: only the step size



Integer model, W = 12, L = 32, K = 3, ε = 2⁻². Left: broadband noise; right: two-tone reference.

$$\mu[n] = r[n] \cdot 2^{-K} / (P + \varepsilon)$$

r[n] ∈ [0.5, 1) is where P + ε sits in its octave

Broadband

r̄ moves smoothly between 0.56 and 0.78. NLMS run with the constant step $\bar{r} \cdot 2^{-K}$ matches NLMS-Shift within 0.27 dB at all 17 levels.

Tonal

P is nearly constant, so r̄ jumps from 0.87 to 0.62 within 0.75 dB of level at −21.1 dBFS: a step change plain NLMS does not have.

Design rule

Choose K and ε for the expected input level; that matters more than the shift itself.

NLMS-Shift costs NLMS's area but LMS's cycles

Complete board design, EP4CE115F29C7	LMS	NLMS	NLMS-Shift
Logic elements	1,792	2,626	2,610
Registers	1,123	1,404	1,186
9-bit multipliers	4	4	4
Fmax, slow 85 °C (MHz)	70.08	59.75	58.66
Setup slack at 50 MHz (ns)	5.731	3.263	2.953
Latency (cycles)	35	37	37
Busy cycles per sample (board)	**73**	**130**	**72**
RTL bit toggles per sample	5,668	7,146	6,012

Quartus Prime Lite 25.1, commit da0b6f4. All three close timing at 50 MHz with no critical warnings.

45% fewer busy cycles than NLMS

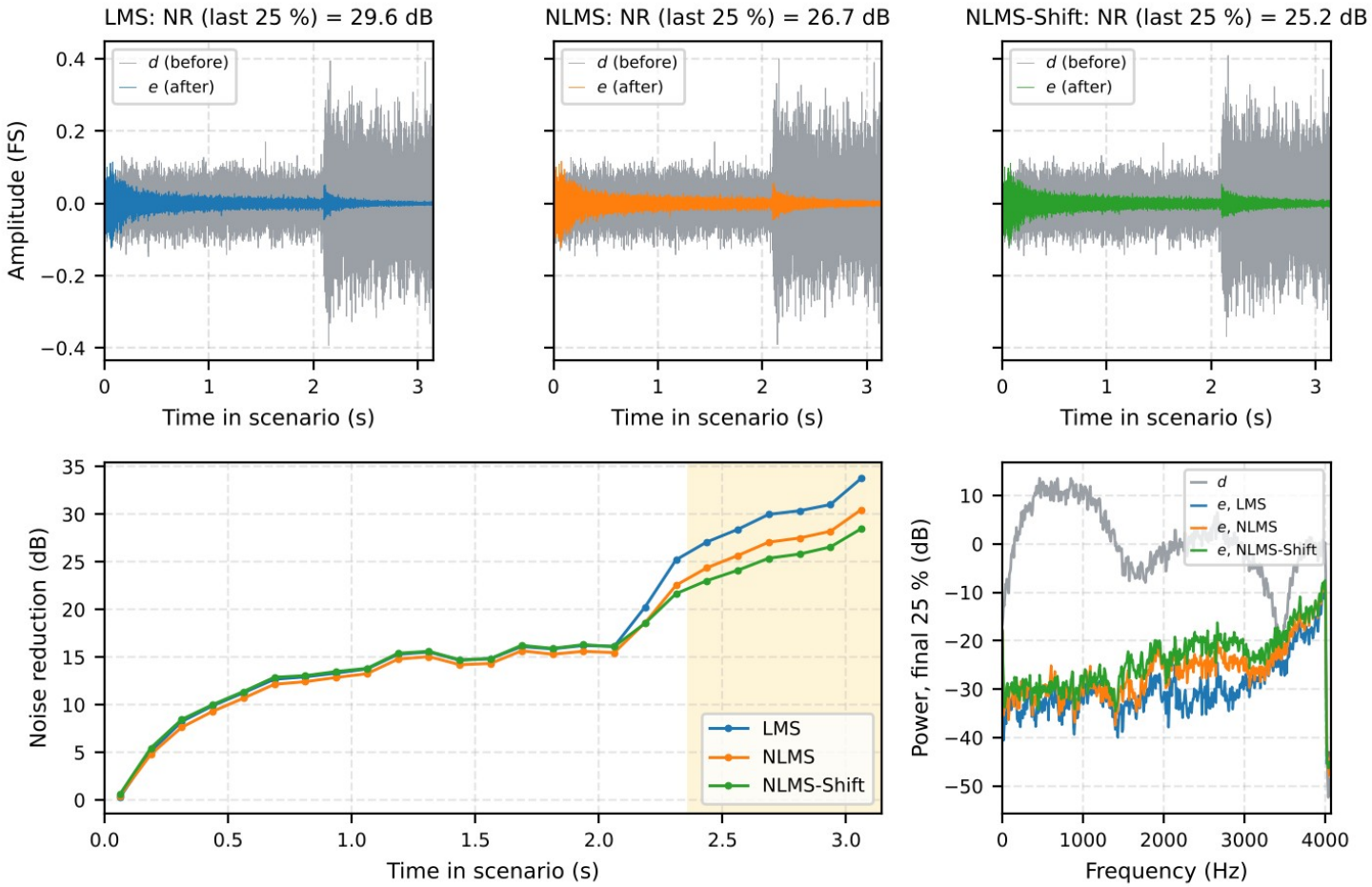
15% fewer registers than NLMS

31% smaller: LMS vs either normalised variant



Power: in progress. Gate-level, activity-based energy per sample is being measured.

On the board, every output sample matches the model



Signal Tap captures of real codec data, amplitude-step noise: d and e (top), running NR and spectra (bottom).

25

Signal Tap captures, 3 algorithms, 2 sessions

724,459

output samples compared with the integer model

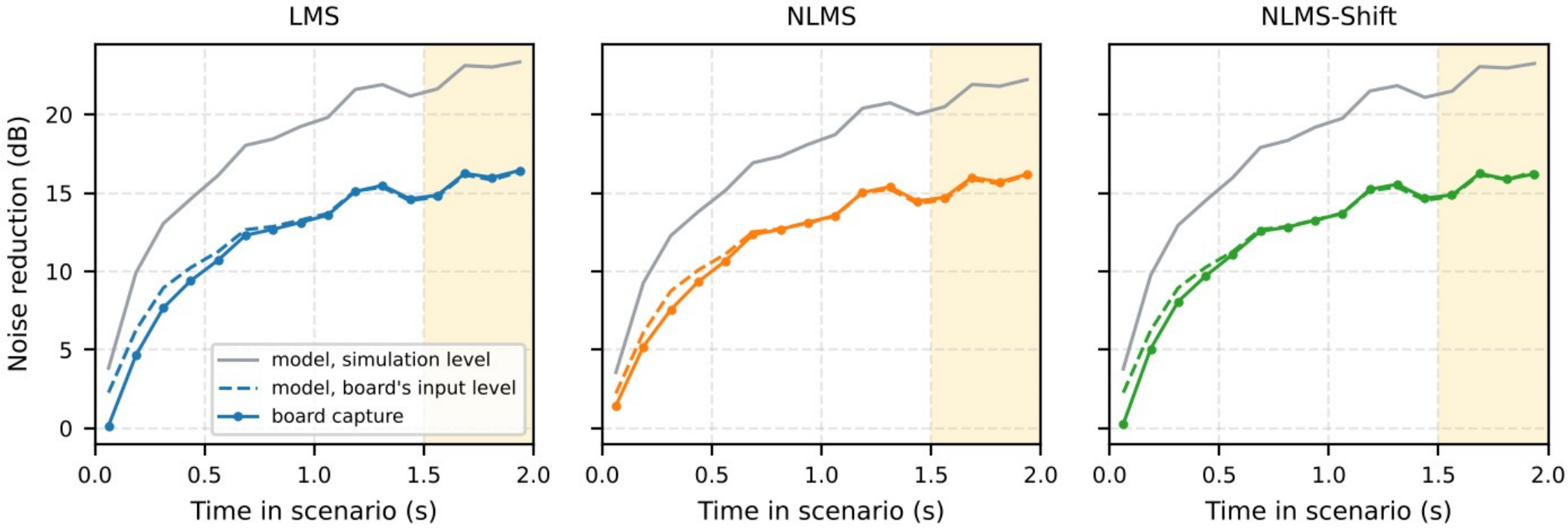
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mismatches

GPIO timing (LMS / NLMS / Shift)

- 6,144 cycles per sample = 8,138.02 Hz
- in → out: 36 / 38 / 38 cycles
- busy: 73 / 130 / 72 cycles, as in RTL

Board NR equals the model at the board's input level



Stationary, final NR (dB)	Board	Model, sim level	Model, board level
LMS	**15.88**	22.77	**15.76**
NLMS	**15.65**	21.60	**15.53**
NLMS-Shift	**15.80**	22.68	**15.78**

Why lower than simulation? The codec delivered x and d at about −27 dBFS, 3 dB below the simulation level, so adaptation is slower. Level-matched, the model agrees within 0.2 dB (stationary, path switch).

Freeze, DAC word and a 10-minute soak all pass

Freeze switch

PASS

SW[1] flipped 1.44 s into the scenario. Output is bit-exact with adaptation stopped from that sample.

NR 15.4 → 15.2 dB frozen (15.9 dB if it had kept adapting)

DAC word, ANC mode

PASS

With SW[0] = 0, the word sent to the DAC equals {e, 0000}.

1,792 / 1,792 samples match

DAC word, bypass

PASS

With SW[0] = 1, the top 12 bits of the DAC word equal d.

1,792 / 1,792 samples match

10-minute soak

PASS

Looped noise for 10 minutes; timing unchanged and no saturation, NACK or overrun flag.

6,144 cycles/sample · 38 latency · 72 busy

The analogue LINE OUT could not be recorded with the equipment available, so the proof stops at the DAC interface.

How ULLAP sits among related work

Work	Platform	Normaliser	Reported
Garrett et al. 2002	ASIC, NLMS equaliser	Power-of-two (leading-zero count)	Origin of the shift idea
Zhao et al. 2018	Virtex-7, beamformer	Quantised normaliser, CSD	≥21% fewer LUTs vs divider
Huang et al. 2024	DE2-115, 16-tap FIR	4-bit LUT reciprocal	10.52 vs 18.60 nJ/sample
Huang & Yao 2026	DE2-115, 5th-order IIR	LOD + shift + 2-segment reciprocal	5.07 vs 8.33 nJ/sample
Rodrigo & Buddhika 2024 (base)	MATLAB / Vivado sim	LMS only	No numeric results
ULLAP (this work)	**DE2-115 + WM8731, 32-tap**	**LMS | divider | shift, one core**	**Equal tuning, 20-seed CI, bit-exact on board**

Not a head-to-head benchmark. Platforms, word lengths and tasks differ. What ULLAP adds is three variants on one core under equal tuning, with a codec and a bit-exact chain to the board. No energy comparison until gate-level power is done.

Limitations, stated plainly

Not better than NLMS

NLMS-Shift matches NLMS in cancellation; it is NLMS with a level-dependent step (×0.5 to ×1). The idea is prior art.

Simulated acoustic paths

Paths are simulated even for real recordings. The base paper's closed-loop feedback (howling) is not covered.

No analogue output recording

Proof stops at the DAC word; LINE OUT was not recorded.

Level-dependent board NR

14–17 dB on the board vs 19–28 dB in simulation, because the input was 3 dB lower. Set codec gain or retune μ.

Power not final

Only vectorless estimates (upper bounds). Gate-level energy per sample is in progress.

Speech limits adaptation

Freezing during speech with a voice-activity detector is not evaluated.

Conclusion

- One serial 12-bit, 32-tap core runs LMS, NLMS and NLMS-Shift on a low-cost Cyclone IV board, with no processor.
- Under equal tuning, NLMS-Shift matches NLMS within the 95% CI using 72 instead of 130 busy cycles per sample.
- Real noise: 11.7–14.9 dB removed, 18.5–31.9 dB without speech, so speech is the limit.
- On the board, 724,459 samples are bit-exact; board NR equals the model within 0.2 dB at the received level.

Future work

- Gate-level energy per sample, then compare with Huang & Yao
- Voice-activity detector driving the freeze input
- Codec gain or step retune for the expected level
- Multi-channel time-sharing (core is ~98% idle)
- Public release with a DOI

Thank you · Questions?